Week 05 Reading: Peaks over threshold

The GPD, choosing a threshold, and what to do about clustered exceedances

2026-09-21

Submit your questions on Canvas by Sunday 9/20 at 11:59 pm, before Monday's class.

Assigned

1. Chapter 4 of Coles (2001), "Threshold Models", pages 74–84.
   - Section 4.1 (p. 74): useful introduction
   - Section 4.2 (pp. 75–78): follow the argument, it's OK if you skim the math
   - Section 4.3 (pp. 78–84): most important. Threshold choice in 4.3.1, parameter estimation in 4.3.2, and return levels in 4.3.3.
   - Section 4.4: optional examples. If you read any of it, read the daily rainfall example in 4.4.1
2. Section 5.3 of Coles (2001), "Modeling Stationary Series", pages 97–100.
3. Part IV of the HEC "Extreme Value Theory" series (slides 34–48), which you may have already watched for week 4. It covers the GPD and peaks over threshold at a slower pace than Coles. The playlist and the slides with speaker notes are the same sources as last week.

Bibliography

Coles, S. (2001). *An Introduction to Statistical Modeling of Extreme Values*. London: Springer.